Module III

Part -II - Bitcoin

Bitcoin

First application of blockchain technology.

Very first fully decentralized digital currency.

The true identity of the Bitcoin inventor is unknown, the name Satoshi Nakamoto is believed to be a seudonym.

In 2008, Bitcoin was introduced through a paper called, Bitcoin: A Peer-to-Peer Electronic Cash System, written by Satoshi Nakamoto. (https://bitcoin.org/bitcoin.pdf)

Bitcoin definition

It is a combination of peer-to-peer network, protocols, software that facilitate the creation and usage of the digital currency named bitcoin. Nodes in this peer-to-peer network talk to each other using the Bitcoin protocol.

Bitcoin with a capital B - Bitcoin protocol,

bitcoin with a lowercase b - bitcoin currency.

Elements of Bitcoin

1. Digital keys
2. Addresses
3. Transactions
4. Blockchain
5. Miners
6. The Bitcoin network
7. Wallets (client software)

example :- how a payment can be sent from one user to another on the Bitcoin network.

Assume that the parties are using Blockchain wallet for mobile devices.

Step 1:

First, either the payment is requested from a user by sending his Bitcoin address to the sender via email or some other means.

The sender can also initiate a transfer to send money to another user.

In both cases, the address of beneficiary is required.

bitcoin payment request (using Blockchain wallet) ->

Step 2:

The sender either enters the receiver's address or scans the QR code that has the Bitcoin address, amount and optional description encoded in it.

The wallet application recognizes this QR code and decodes it into something like,

```
Please send <Amount> BTC to the Bitcoin address <receiver's Bitcoin address>.
```

Please send 0.00033324 BTC to the Bitcoin address 1JzouJCVmMQBmTcd8K4Y5BP36gEFNn1ZJ3.

bitcoin://1JzouJCVmMQBmTcd8K4Y5BP36gEFNn1ZJ3?amount=0.00033324

Step 3:

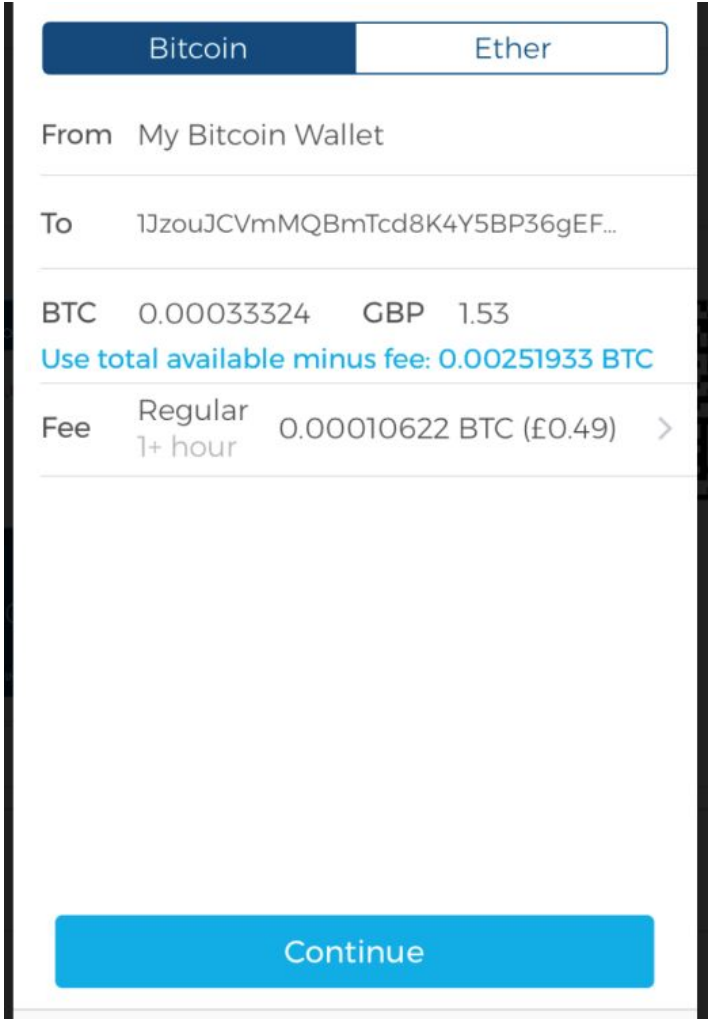
In the wallet application of the sender, this transaction is constructed by following some rules and broadcasted to the Bitcoin network.

This transaction is digitally signed using the private key of the sender before broadcasting it.

Fee is calculated based on the size of the transaction and a fee rate is a value that depends on the volume of the transaction in the network.

This is represented in Satoshis/byte.

Fee in Bitcoin network ensures that your transaction will be included by miners in the block.

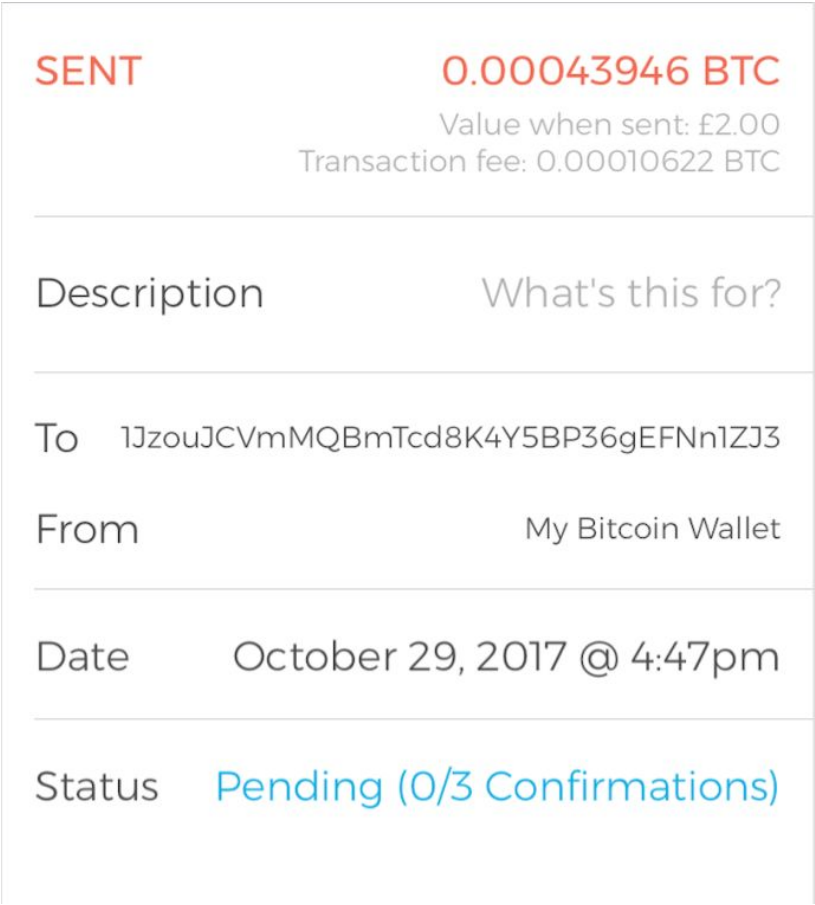


The fee is usually estimated and calculated by the Bitcoin wallet software automatically before sending the transactions.

The higher the transaction fee the more chances are that your transaction will be picked up at priority and included in the block.

This task is performed by the miners.

Once the transaction is sent it will appear as shown here in the Blockchain wallet software:



Step 4:

At this stage, the transaction has been constructed, signed and sent out to the Bitcoin network.

This transaction will be picked up by miners to be verified and included in the block.

Confirmation is pending for this transaction.

These confirmations will start to appear as soon as the transaction is verified, included in the block, and mined.

Also, the appropriate fee will be deducted from the original value to be transferred and will be paid to the miner who has included it in the block for mining.

Bitcoin transactions are serialized for transmission over the network and encoded in hexadecimal format.

The payment transaction in the Bitcoin network can be divided into the following steps:

1. Transaction starts with a sender signing the transaction with their private key
2. Transaction is serialized so that it can be transmitted over the network
3. Transaction is broadcasted to the network
4. Miners listening for the transactions picks up the transaction
5. Transaction are verified for their validity by the miners
6. Transaction are added to the candidate/proposed block for mining
7. Once mined, the result is broadcasted to all nodes on the Bitcoin network

Bitcoin Denominations

DENOMINATION	ABBREVIATION	FAMILIAR NAME	VALUE IN BTC
Satoshi	SAT	Satoshi	0.00000001 BTC
Microbit	µBTC (uBTC)	Microbitcoin or Bit	0.000001 BTC
Millibit	mBTC	Millibitcoin	0.001 BTC
Centibit	cBTC	Centibitcoin	0.01 BTC
Decibit	dBTC	Decibitcoin	0.1 BTC
Bitcoin	BTC	Bitcoin	1 BTC
DecaBit	daBTC	Decabitcoin	10 BTC
Hectobit	hBTC	Hectobitcoin	100 BTC
Kilobit	kBTC	Kilobitcoin	1000 BTC
Megabit	MBTC	Megabitcoin	1000000 BTC

Digital keys and addresses

On the Bitcoin network, possession of bitcoins and transfer of value via transactions is reliant upon private keys, public keys, and addresses.

Elliptic Curve Cryptography (ECC) is used to generate public and private key pairs in the Bitcoin network.

Private keys in Bitcoin

Private keys are required to be kept safe and normally resides only on the owner's side.

Private keys are used to digitally sign the transactions proving the ownership of the bitcoins.

Private keys are fundamentally 256-bit numbers randomly chosen in the range specified by the *secp256k1* ECDSA curve recommendation

Private keys are usually encoded using Wallet Import Format (WIF) in order to make them easier to copy and use.

It is a way to represent the full size private key in a different format.

WIF can be converted into a private key and vice versa.

Private key:
A3ED7EC8A03667180D01FB4251A546C2B9F2FE33507C68B7D9D4E1FA5714195201

in WIF format:
L2iN7umV7kbr6LuCmgM27rBnptGbDVc8g4ZBm6EbgTPQXnj1RCZP

(https://gobittest.appspot.com/PrivateKey)

A mini private key format is sometimes used to create the private key with a maximum of up to 30 characters in order to allow storage where physical space is limited.

The private key encoded using mini private key format is also sometimes called minikey.

A mini private key can be converted into a normal size private key but an existing normal size private key cannot be converted into a mini private key.

Public keys in Bitcoin

All network participants can see it.

Public keys are derived from private keys.

Once a transaction signed with the private key is broadcasted on the Bitcoin network, public keys are used by the nodes to verify that the transaction has indeed been signed with the corresponding private key.

This process of verification proves the ownership of the bitcoin.

Bitcoin uses ECC based on the *secp256k1* standard

A public key is 256-bits in length.

Uncompressed public keys use 0x04 as the prefix.

Addresses in Bitcoin

A bitcoin address is created by taking the corresponding public key of a private key and hashing it twice, first with the SHA-256 algorithm and then with RIPEMD-160.

The resultant 160-bit hash is then prefixed with a version number and finally encoded with a Base58Check encoding scheme.

The bitcoin addresses are 26-35 characters long and begin with digit 1 or 3.

Eg: 1ANAguGG8bikEv2fYsTBnRUmx7QUcK58wt

This is also commonly encoded in a QR code for easy distribution.

There are two types of addresses, the commonly used P2PKH and another P2SH type, starting with number 1 and 3, respectively.

Base58Check encoding

Bitcoin addresses are encoded using the Base58Check encoding.

This encoding is used to limit the confusion between various characters, such as 0OIl as they can look the same in different fonts.

The encoding basically takes the binary byte arrays and converts them into human-readable strings.

This string is composed by utilizing a set of 58 alphanumeric symbols.

Characters include, 1-9 (no 0), A-Z (without O and I), a-z (without l)

Transactions

Transactions can be as simple as just sending some bitcoins to a bitcoin address.

Each transaction is composed of at least one input and output.

Inputs can be thought of as coins being spent that have been created in a previous transaction and outputs as coins being created.

If a transaction is minting new coins, then there is no input and therefore no signature is needed.

If a transaction is to send coins to some other user (a bitcoin address), then it needs to be signed by the sender with their private key and a reference is also required to the previous transaction in order to show the origin of the coins.

The transaction life cycle

1. A user/sender sends a transaction using wallet software or some other interface.

2. The wallet software signs the transaction using the sender's private key.

3. The transaction is broadcasted to the Bitcoin network using a flooding algorithm.

4. Mining nodes (miners) who are listening for the transactions verify and include this transaction in the next block to be mined. Just before the transaction are placed in the block they are placed in a special memory buffer called transaction pool.

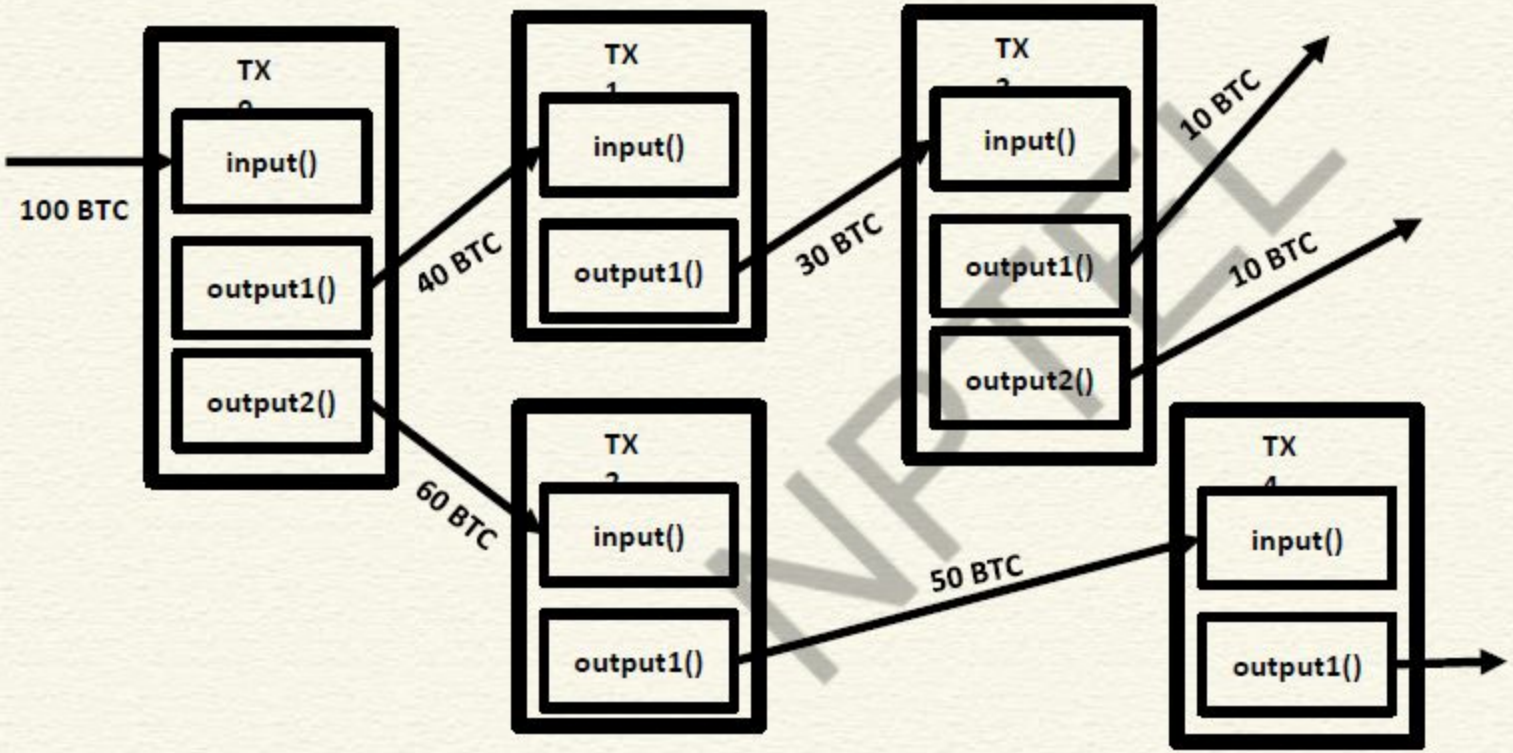
5. Mining starts, which is a process by which the blockchain is secured and new coins are generated as a reward for the miners who spend appropriate computational resources.

6. Once a miner solves the PoW problem it broadcasts the newly mined block to the network.

7. The nodes verify the block and propagate the block further, and confirmations start to generate.

8. Finally, the confirmations start to appear in the receiver's wallet and after approximately three confirmations, the transaction is considered finalized and confirmed. However, three to six is just a recommended number; the transaction can be considered final even after the first confirmation.

Bitcoin Transactions and Input and Output



Field	Size	Description
Version number	4 bytes	Used to specify rules to be used by the miners and nodes for transaction processing.
Input counter	1-9 bytes	The number (positive integer) of inputs included in the transaction.
List of inputs	Variable	Each input is composed of several fields, including `Previous Tx hash`, `Previous Txout-index`, `Txin-script length`, `Txin-script`, and optional sequence number. The first transaction in a block is also called a coinbase transaction. It specifies one or more transaction inputs.
Output counter	1-9 bytes	A positive integer representing the number of outputs.
List of outputs	Variable	Outputs included in the transaction.
Lock time	4 bytes	This field defines the earliest time when a transaction becomes valid. It is either a Unix timestamp or block height.

Transaction data Structure in Bitcoin

Transaction fee

Transaction fees are charged by the miners.

The fee charged is dependent upon the size and weight of the transaction.

The fees are used as an incentive for miners to encourage them to include a user transaction in the block the miners are creating.

All transactions end up in the memory pool, from where miners pick up transactions based on their priority to include them in the proposed block.

A transaction with a higher fee will be picked up sooner by the miners.

The script language

Bitcoin uses a simple stack-based language called script to describe how bitcoins can be spent and transferred.

It is not Turing complete and has no loops.

It is evaluated from the left to the right using a Last In, First Out (LIFO) stack.

Scripts use various opcodes or instructions to define their operation.

Coinbase transactions

A coinbase transaction or generation transaction is always created by a miner and is the first transaction in a block.

It is used to create new coins.

It includes a special field, also called coinbase, which acts as an input to the coinbase transaction.

This transaction also allows up to 100 bytes of arbitrary data that can be used to store arbitrary data.

There is a special condition for coinbase transactions.

It cannot be spent until at least 100 blocks in order to avoid a situation where the block may be declared stale later on.

In the genesis block, this transaction included the most famous comment taken from The Times newspaper:

"The Times 03/Jan/2009 Chancellor on brink of second bailout for banks."

This message is a proof that the genesis block was not mined earlier than January 3, 2009.

First Bitcoin block (genesis block) was created on January 3, 2009 and this news excerpt was taken from that day's newspaper.

A coinbase transaction input has the same number of fields as usual transaction input.

But the structure contains coinbase data size and coinbase data fields instead of unlocking script size and unlocking script fields.

Also, it does not have a reference pointer to the previous transaction.

Field	Size	Description
Transaction hash	32 bytes	Set to all zeroes as no hash reference is used
Output index	4 bytes	Set to `0xFFFFFFFF`
Coinbase data length	1-9 bytes	2 bytes-100 bytes
Data	Variable	Any data
Sequence number	4 bytes	Set to `0xFFFFFFFF`

Coinbase transaction structure

Transaction verification

This verification(Rule checking) process is performed by Bitcoin nodes.

1. Check the syntax and ensure that the syntax and data structure of the transaction conforms to the rules provided by the protocol.

2. Verify that no transaction inputs and outputs are empty.

3. Check whether the size in bytes is less than the maximum block size.

4. The output value must be in the allowed money range (0 to 21 million BTC).

5. All inputs must have a specified previous output, except for coinbase transactions.

6. Verify that nLockTime must not exceed 31-bits. (nLockTime specifies the time before which transaction will not be included in the block.)

7. For a transaction to be valid, it should not be less than 100 bytes.

8. The number of signature operations in a standard transaction should be less than or not more than two.

9. Reject nonstandard transactions;

The `isStandard()` checks specify that only standard transactions are allowed.

10. A transaction is rejected if there is already a matching transaction in the pool or in a block in the main branch.

11. The transaction will be rejected if the referenced output for each input exists in any other transaction in the pool.

12. For each input, there must exist a referenced output unspent transaction.

13. For each input, if the referenced output transaction is the coinbase, it must have at least 100 confirmations; otherwise, the transaction will be rejected.

14. For each input, if the referenced output does not exist or has been spent already, the transaction will be rejected.

15. Using the referenced output transactions to get input values, verify that each input value, as well as the sum, is in the allowed range of 0-21 million BTC. Reject the transaction if the sum of input values is less than the sum of output values.

16. Reject the transaction if the transaction fee would be too low to get into an empty block.

17. Each input unlocking script must have corresponding valid output scripts.

Blockchain

Blockchain is a public ledger of a timestamped, ordered, and immutable list of all transactions on the Bitcoin network.

Each block is identified by a hash in the chain and is linked to its previous block by referencing the previous block's hash.

Field	Size	Description
Block size	4 bytes	This is the size of the block.
Block header	80 bytes	This includes fields from the block header
Transaction counter	Variable	This field contains the total number of transactions in the block, including the coinbase transaction. Size ranges from 1-9 bytes
Transactions	Variable	All transactions in the block.

The structure of a block header

Field	Size	Description
Version	4 bytes	The block version number that dictates the block validation rules to follow.
Previous block's header hash	32 bytes	This is a double SHA-256 hash of the previous block's header.
Merkle root hash	32 bytes	This is a double SHA-256 hash of the Merkle tree of all transactions included in the block.
Timestamp	4 bytes	This field contains the approximate creation time of the block in the Unix epoch time format. More precisely, this is the time when the miner has started hashing the header. (The time from the miner's point of view.)
Difficulty target	4 bytes	This is the current difficulty target of the network/block.
Nonce	4 bytes	This is an arbitrary number that miners change repeatedly to produce a hash that is lower than the difficulty target.

The genesis block

The first block is not linked to any previous block and is known as the genesis block.

The genesis block was hardcoded in the bitcoin core software.

```
static CBlock CreateGenesisBlock(uint32_t nTime, uint32_t nNonce, uint32_t
nBits, int32_t nVersion, const CAmount& genesisReward)
{
   const char* pszTimestamp = "The Times 03/Jan/2009 Chancellor on brink of
second bailout for banks";
   const CScript genesisOutputScript = CScript() <<
ParseHex("04678afdb0fe5548271967f1a67130b7105cd6a828e03909a67962e0ea1f61deb
649f6bc3f4cef38c4f35504e51ec112de5c384df7ba0b8d578a4c702b6bf11d5f") <<
OP_CHECKSIG;
   return CreateGenesisBlock(pszTimestamp, genesisOutputScript, nTime,
nNonce,
   nBits, nVersion, genesisReward);
}
```

Bitcoin provides protection against double spending by enforcing strict rules on transaction verification and via mining.

Block height is the number of blocks before a particular block in the blockchain.

Current height: 838,586 at 2024-04-10 14:41:33

Blocks List The total Number of 838,587 Blocks

2024-03-10 → 2024-04-10 Export

Height	Relayed By	Time	Tx Count	Reward (BTC)	Size (KB)	Fees (BTC)	Volume (BTC)
838,586	ViaBTC	2024-04-10 14:41:33	4,159	6.49281097	1,660.39	0.24281097	2,727.63107297
838,585	Foundry USA	2024-04-10 14:32:14	4,161	6.38756983	1,780.19	0.13756983	1,209.41989031
838,584	SBI Crypto	2024-04-10 14:31:12	3,979	6.50008779	1,560.99	0.25008779	3,485.79469659

https://explorer.btc.com/btc/blocks

Mining

Mining is a process by which new blocks are added to the blockchain.

Blocks contain transactions that are validated via the mining process by mining nodes on the Bitcoin network.

Blocks, once mined and verified are added to the blockchain which keeps the blockchain growing.

This process is resource-intensive due to the requirements of PoW where miners compete in order to find a number which is less than the difficulty target of the network.

This difficulty in finding the correct value (also called sometimes the mathematical puzzle) is there to ensure that the required resources have been spent by miners before a new proposed block can be accepted.

New coins are minted by the miners by solving the PoW problem, also known as partial hash inversion problem.

This process consumes a high amount of resources including computing power and electricity.

This process also secures the system against frauds and double spending attacks while adding more virtual currency to the Bitcoin ecosystem.

Roughly one new block is created (mined) every 10 minutes.

This frequency needs to be maintained by the Bitcoin network.

Miners are rewarded with new coins if and when they discover new blocks by solving PoW.

The rate of creation of new bitcoins decreases by 50%, every 210,000 blocks, roughly every 4 years.

Miners are paid transaction fees in return for including transactions in their proposed blocks.

New blocks are created at an approximate fixed rate of every 10 minutes.

When bitcoin was initially introduced, the block reward was 50 bitcoins; then in 2012, this was reduced to 25 bitcoins.

In July 2016, this was further reduced to 12.5 coins (12 coins), From 12 June 2020, reward was further reduced to approximately six (6.25) coins.

This mechanism is hardcoded in Bitcoin to regulate, control inflation and limit the supply of bitcoins.

Approximately 144 blocks, that is, 864 bitcoins are generated per day (now).

The number of actual coins can vary per day; however, the number of blocks remains at 144 per day.

Bitcoin supply is also limited and in 2140, almost 21 million bitcoins will be finally created and no new bitcoins can be created after that.

Bitcoin miners, however, will still be able to profit from the ecosystem by charging transaction fees.

Tasks of the miners

1. Synching up with the network: Once a new node joins the bitcoin network, it downloads the blockchain by requesting historical blocks from other nodes.

2. Transaction validation: Transactions broadcasted on the network are validated by full nodes by verifying and validating signatures and outputs.

3. Block validation: Miners and full nodes can start validating blocks received by them by evaluating them against certain rules. This includes the verification of each transaction in the block along with verification of the nonce value.

4. Create a new block: Miners propose a new block by combining transactions broadcasted on the network after validating them.

5. Perform Proof of Work: This task is the core of the mining process and this is where miners find a valid block by solving a computational puzzle.

The block header contains a 32-bit nonce field and miners are required to repeatedly vary the nonce until the resultant hash is less than a predetermined target.

6. Fetch reward: Once a node solves the hash puzzle (PoW), it immediately broadcasts the results, and other nodes verify it and accept the block.

There is a slight chance that the newly minted block will not be accepted by other miners on the network due to a clash with another block found at roughly the same time, but once accepted, the miner is rewarded with 6 bitcoins and any associated transaction fees.

Proof of Work (PoW)

This is a proof that enough computational resources have been spent in order to build a valid block.

PoW is based on the idea that a random node is selected every time to create a new block.

In this model, nodes compete with each other in order to be selected in proportion to their computing capacity.

The following equation sums up the PoW requirement in bitcoin:

```
H ( N || P_hash || Tx || Tx || . . . Tx) < Target
```

Where N is a nonce, P_hash is a hash of the previous block, Tx represents transactions in the block, and Target is the target network difficulty value.

The only way to find this nonce is the brute force method.

The mining algorithm

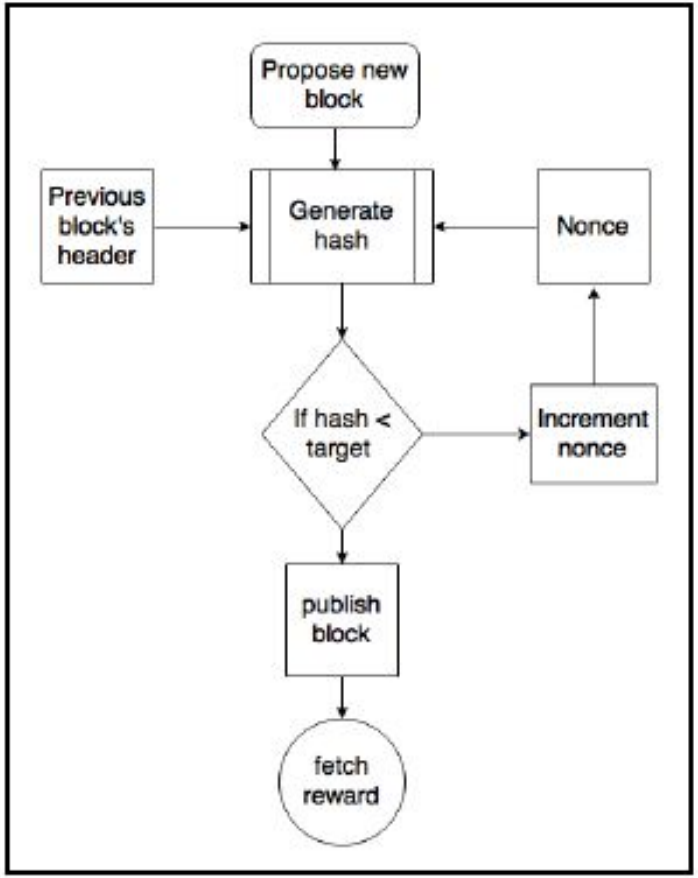
1. The previous block's header is retrieved from the bitcoin network.

2. Assemble a set of transactions broadcasted on the network into a block to be proposed.

3. Compute the double hash of the previous block's header combined with a nonce and the newly proposed block using the SHA-256 algorithm.

4. Check if the resultant hash is lower than the current difficulty level (target) then PoW is solved. As a result of successful PoW the discovered block is broadcasted to the network and miners fetch the reward.

5. If the resultant hash is not less than the current difficulty level (target), then repeat the process after incrementing the nonce.



Mining process

Mining difficulty increased over time and bitcoins that could be mined by single CPU laptop computers now require dedicated mining centers to solve the hash puzzle.

The hash rate

The hashing rate represents the rate of calculating hashes per second.

This is the speed at which miners in the Bitcoin network are calculating hashes to find a block.

In early days of bitcoin, it used to be quite small as CPUs were used.

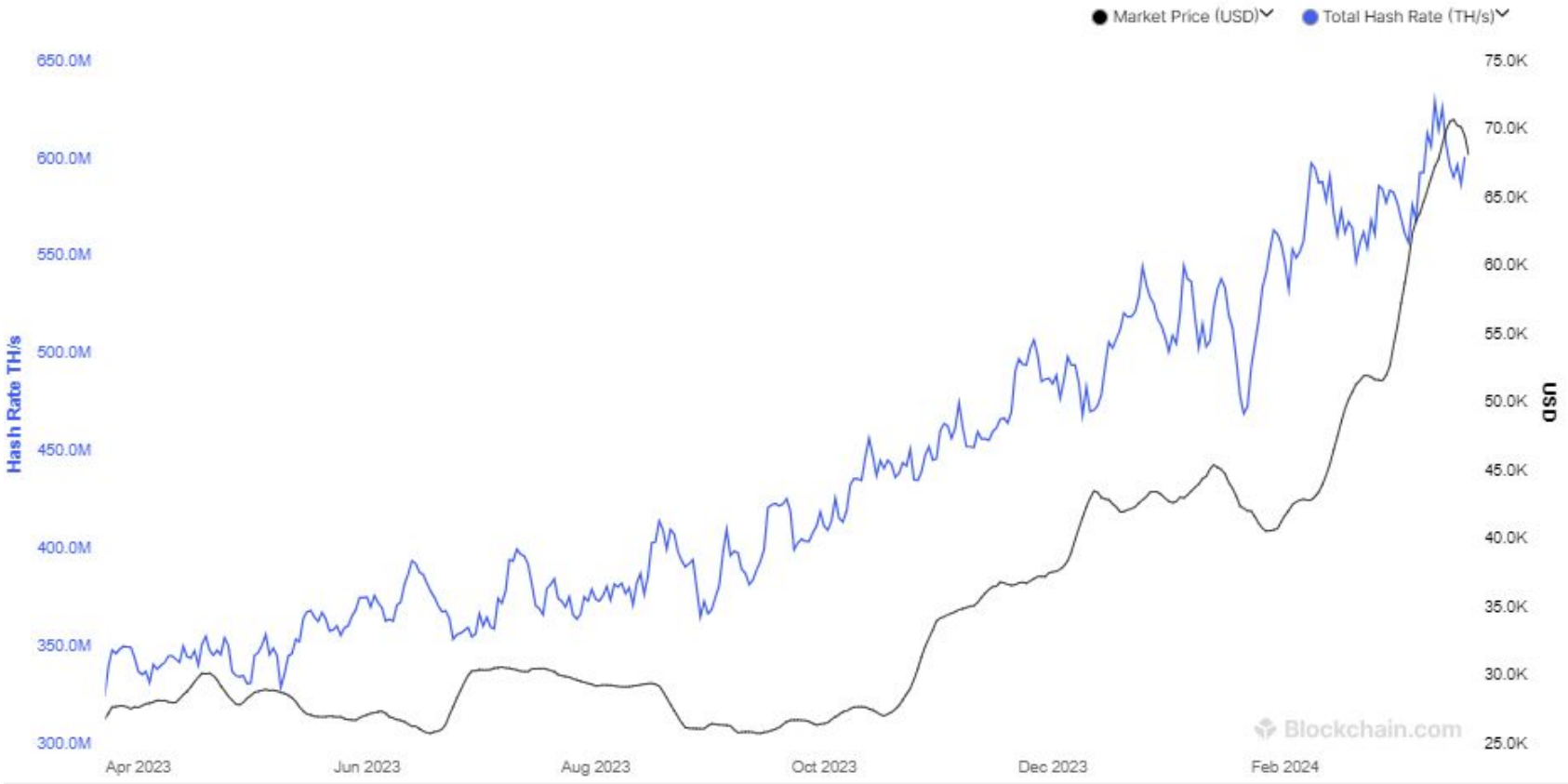
However, with dedicated mining pools and ASICs now, this has gone up exponentially in the last few years.

This has resulted in increased difficulty of the Bitcoin network.

Hash rate is measured in Tera hashes per second (TH/s) or Exa Hashes/s (EH/s)

Current hash rate is around 601,016,901 TH/s. (20 March 2024)

This means that in 1 second, the Bitcoin network miners are computing more than 601,016,901,000,000,000,000 hashes.



Explanation

Data

Hash Rate over time

Wallets

The wallet software is used to store private or public keys and Bitcoin address.

It performs various functions, such as receiving and sending bitcoins.

Nowadays, software usually offers both functionalities: Bitcoin client and wallet.

On the disk, the Bitcoin core client wallets are stored as the Berkeley DB file.

Private keys are generated by randomly choosing a 256-bit number by wallet software.

Private keys are used by wallets to sign the outgoing transactions.

Wallets do not store any coins, and there is no concept of wallets storing balance or coins for a user.

In the Bitcoin network, coins do not exist; instead, only transaction information is stored on the blockchain, which are then used to calculate the number of bitcoins.

In Bitcoin, there are different types of wallets that can be used to store private keys.

As a software program, they also provide some functions to the users to manage and carry out transactions on the Bitcoin network.

1. Non-deterministic wallets

These wallets contain randomly generated private keys and are also called just a bunch of key wallets.

The Bitcoin core client generates some keys when first started and generates keys as and when required. There is a need to create regular backups of the keys and protect them appropriately, for example, by encrypting them in order to prevent theft or loss.

2. Deterministic wallets

In this type of wallet, keys are derived out of a seed value via hash functions.

This seed number is generated randomly and is commonly represented by human-readable mnemonic code words.

Mnemonic code words are defined in BIP 39, a Bitcoin improvement proposal for mnemonic code for generating deterministic keys.

3. Hierarchical Deterministic wallets

Hierarchical Deterministic (HD) wallets store keys in a tree structure derived from a seed.

Defined in protocols BIP32 and BIP44.

The seed generates the parent key (master key), which is used to generate child keys and, subsequently, grandchild keys.

Key generation in HD wallets does not generate keys directly; instead, it produces some information (private key generation information) that can be used to generate a sequence of private keys.

The complete hierarchy of private keys in an HD wallet is easily recoverable if the master private key is known.

It is because of this property that HD wallets are very easy to maintain and are highly portable.

There are many free and commercially available HD wallets available.

For example, Trezor (https://trezor.io), Jaxx (https://jaxx.io/) and Electrum (https://electrum.org/).

4. Brain wallets

The key idea is that a passphrase is used to derive the private key.

This master key is used in HD wallets.

This can result in a full HD wallet that is derived from a single memorized password.

This is known as a brain wallet.

This method is prone to password guessing and brute force attacks but techniques such as key stretching can be used to slow down the progress made by the attacker.

5. Paper wallets

As the name implies, this is a paper-based wallet with the required key material printed on it.

It requires physical security to be stored.

Paper wallets can be generated online from various service providers, such as

https://bitcoinpaperwallet.com/ or

https://www.bitaddress.org/.

Paper wallet

6. Hardware wallets

Usea tamper-resistant device to store keys.

This tamper-resistant device can be custom-built or with the advent of NFC-enabled phones, this can also be a

Secure Element (SE) in NFC phones.

Trezor and Ledger wallets are the most commonly used Bitcoin hardware wallets.

7. Online wallets

Online wallets are stored entirely online and are provided as a service usually via the cloud.

They provide a web interface to the users to manage their wallets and perform various functions such as making and receiving payments.

They are easy to use but imply that the user trusts the online wallet service provider.

An example of online wallet is GreenAddress, https://greenaddress.it/en/.

8. Mobile wallets

Mobile wallets are installed on mobile devices.

They can provide various methods to make payments.

Have the ability to use smartphone cameras to scan QR codes quickly and make payments.

Mobile wallets are available for the Android platform and iOS, for example, Blockchain, breadwallet, Copay, and Jaxx.